Performance Enhancement of Conventional Design of 4-Bit Carry Look-Ahead Adder

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***Abstract*—This research aims to increase performance parameters of conventional CMOS based 4-bit carry look-ahead (CLA) adder. CLA adder in conventional design fully utilize static CMOS (S-CMOS) logic for which its transistor count is quite high. Moreover, due to high input impedance, its delay is also high. To enhance performance and to reduce transistor count, (1) full swing gate diffusion input (GDI) based AND gates and (2) GDI based XOR gates have been utilized rather than using S-CMOS based AND and XOR gates. According to simulation conducted using Cadence tools, the proposed CLA adder design achieved noteworthy improvements in performance compared to the fully S-CMOS based existing design.**

Keywords—adder, carry look-ahead, AND gate, XOR gate, gate diffusion input.

I. INTRODUCTION

The role of binary adder in Arithmetic Logic Units (ALU) is quite crucial since many arithmetic functions and operations need addition of binary numbers. Therefore, adder is considered as a vital component in ALUs [1]. Due to intense importance and application of adders in various operations, the performance of adder needs to be optimized to bring about overall performance enhancement in ALU. Hence, researchers are constantly working on implementing new adder topologies which resulted in designing numerous sorts of adders based on various logic styles [2].

Ripple carry adder (RCA) is considered as the most basic and simplest type of adder [3]. However, due to having long carry chain, delay of RCA is quite high which makes it inappropriate for using in modern high-performance microprocessors. In order to reduce long carry chain for wide adders, the concept of parallel-prefix adders evolved (PPA) [4]. Design of various PPA have been developed and each PPA design retain its advantages and disadvantages [5]. CLA adder is one of the widely used adder topologies [6]. However, the conventional design of CLA adder use S-CMOS logic for which its transistor count is quite high. This high transistor count corresponds to high area consumption in chip. Moreover, it results in very high-power dissipation [7]. In addition, high input impedance of conventional design corresponds to increased delay which makes the operation of adding binary numbers slow [8]. In spite of having these limitations, several factors such as driving capability, robustness against voltage scaling, robustness against transistor sizing etc. enable designers to implement CLA adder using S-CMOS logic [9].

Using low supply voltage and low clock speed can make a circuit consume less power [10]. But applying low voltage and low clock speed make circuits slow. Hence, designers are faced with barriers which make circuit design difficult.

Circuit using fast logic gates can bring about improvement in processing speed. But, no circuit element in this world is ideal and have their own limitations [11].

The prominent way to enhance performance in modern large-scale digital circuits is to design circuits in an efficient manner. Hence, many logic design techniques have been evolved over the years.

In this paper, a hybrid 4-bit CLA adder using GDI technique and S-CMOS logic is being presented. The design of 4-bit adder is important because it is used as a building block in order to implement wide adders [12, 13]. Therefore, performance enhancement of 4-bit adder will result in increasing performance parameters of wide adders. The proposed design although used GDI technique, advantages of S-CMOS logic was kept intact. GDI technique based AND and XOR gates have been used in the proposed design. It resulted in reduction of transistor count which corresponds to low power and surface area. Moreover, since AND and XOR gates are used in input side of CLA adder, input impedance has been reduced significantly which increased speed of the design. The output circuits (which generate carry terms and sum terms) of the proposed design use S-CMOS logic which have high driving power. In this way, the combination of GDI and S-CMOS logic enhanced performance while keeping the advantages conventional CLA adder design exactly the same.

Organization of the remaining portion of this research article is: section II provides overall details for existing CLA adder design and section III presents the methodologies used in case of designing the proposed CLA adder. Simulation result of existing and proposed circuits are presented in section IV. In section V, a comparative analysis between fully S-CMOS based CLA adder and the proposed Adder is presented. Concluding remarks are listed in section VI.

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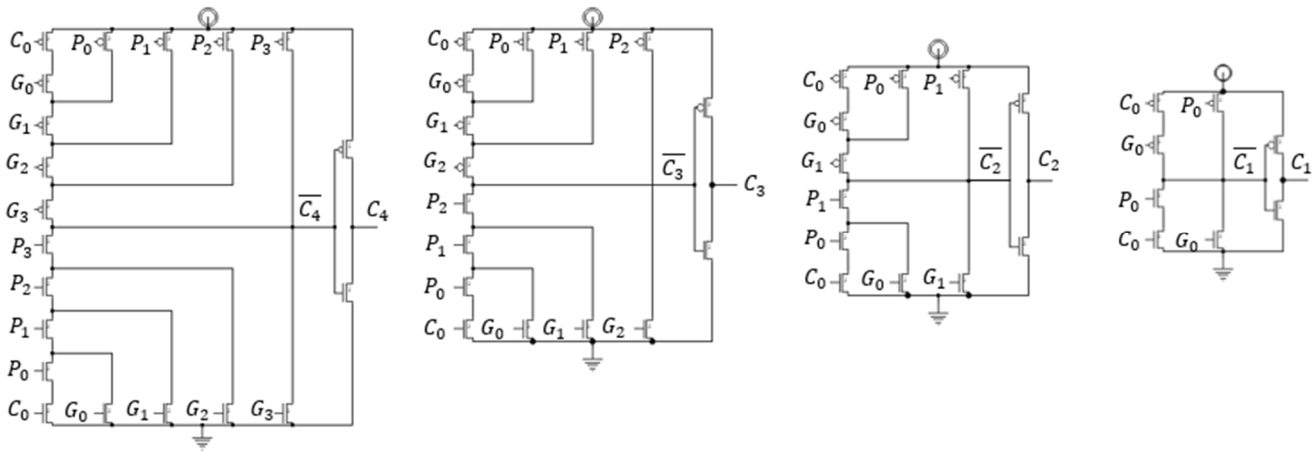


Fig. 1. Circuit diagram for CLA terms.

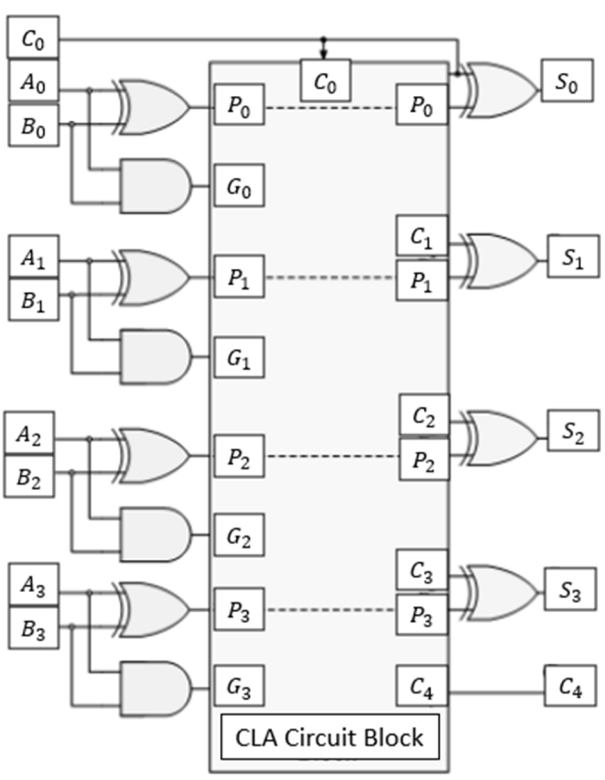


Fig. 2. Block diagram of CLA adder.

II. Literature Review of Conventional CLA Adder

The outputs of a 4-bit CLA adder are represented using the following Boolean equations [6].

$$C_{i+1} = G_i + P_i C_i$$

$$S_i = P_i \oplus C_i$$

Where the term 'i' is an integer number and $0 < i < 3$.

Here the term G_i is called Carry Generate or simply Generate. The term G_i represents AND operation between the ith input bits of the two binary numbers.

$$G_i = A_i B_i$$

On the other hand, P_i is called Carry Propagate or simply Propagate. It represents XOR operation between the ith bits of the two binary numbers.

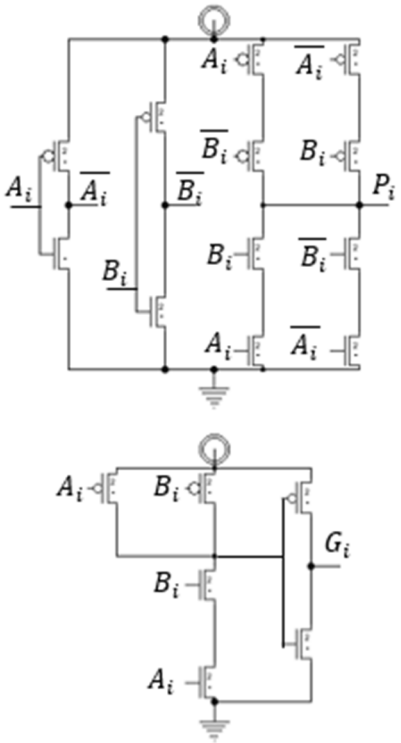


Fig. 3. XOR and AND gate Design used in conventional CLA adder.

$$P_i = A_i \oplus B_i$$

Circuit diagram of CLA terms is represented by fig. 1 and block diagram for CLA adder is represented by fig. 2 [8]. In fig. 3, the AND and XOR gates are presented which are used in conventional CLA adder for generating the propagate and generate signals.

III. Proposed CLA Adder

The proposed design of CLA adder have modifications in the XOR and AND gates which are used for generating propagate and generate terms respectively. The complex CLA circuits which provides outputs of carry bits (C_4, C_3, C_2, and C_1) and the XOR gates for generating outputs of Sum terms (S_3, S_2, S_1, and S_0) are kept same as the existing design. In short, design modifications have been made only in the input side. Output side circuits are kept exactly the same as the conventional design in [6].

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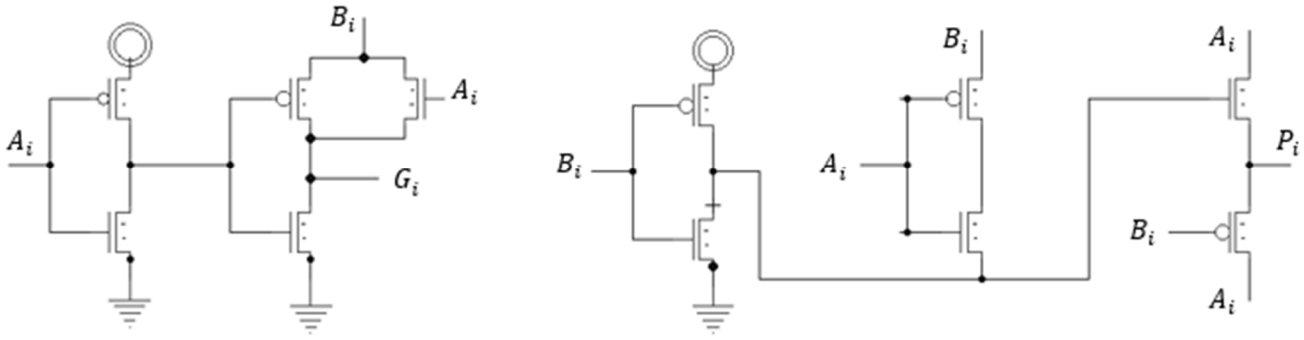


Fig. 4. Proposed AND and XOR gate design for generating G_i and P_i.

A. AND gate design for Generate Terms G_i

In case of designing AND gate in Fig. 4, GDI technique has been used. GDI technique enables designers to implement circuits using very small number of transistors. Therefore, power consumption of such sort of circuits are low [14]. The most basic sort of GDI AND gate only requires 2 transistors [15, 16]. However, this AND gate design has the problem of voltage degradation. GDI AND gate in [17] provides full swing output which means it is not associated with the problem of voltage degradation. AND gate design in [18] requires 5 transistors to implement but have higher delay than AND gate in [17]. Hence, AND gate in [17] has been used to generate G_i signals.

B. XOR gate design for Propagate Terms P_i

Many XOR gate designs have been developed by researchers using different logic styles [19-22]. XNOR gate in [23] can be converted into XOR gate using an inverter in series with the output.

XOR gate in [19-21] provides low logic swing, but require only 4 transistors for implementation. XOR gate in [22] and XNOR gate in [23] uses 6 transistors. XNOR in [23] is required to convert into XOR gate by adding inverter which increases its transistor count and power. XOR gates in [19-22] have more delay. XOR gate in [18] have high power and delay. According to simulation in Cadence, XOR gate in [17] appears to be the most efficient design in case of delay and power. Hence, XOR module in [17] has been chosen for the proposed design. The XOR circuit is shown in Fig. 4.

IV. Simulation Result

In order to analyze the proposed and existing designs, simulations have been conducted in 90 nm technology [24-26]. The software used for simulation purpose is Cadence Virtuoso. For all simulation cases, supply voltage has been set to 1.0 V [27-31]. Simulation result of the CLA adders (proposed and existing) and performance improvement achieved by the proposed design is listed in Table I. The table consists simulation on delay, power and power delay product (PDP). Also, the performance parameters are shown graphically using Fig. 5, 6 and 7.

TABLE I. Simulation Result

CLA Adder	Performance Parameter	Simulation result	Improvement
Existing [6]	Power (μW)	77.338	35.38%
Proposed		49.974	
Existing [6]	Delay (ns)	0.167	14.371%
Proposed		0.143	
Existing [6]	PDP (fJ)	12.915	44.67%
Proposed		7.146	

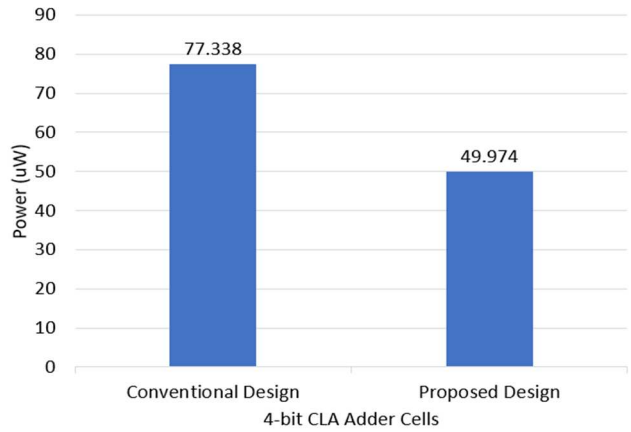


Fig. 5. Average power comparison.

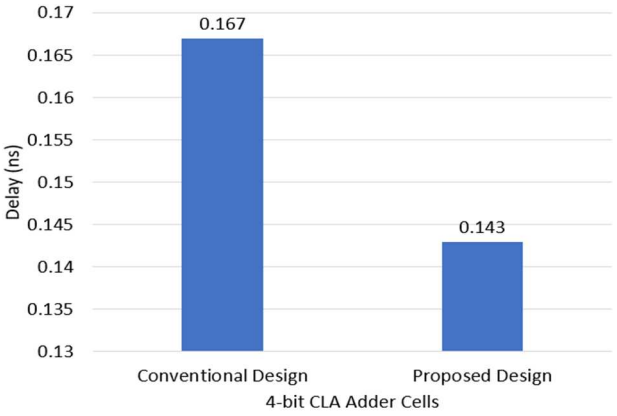


Fig. 6. Propagation delay comparison.

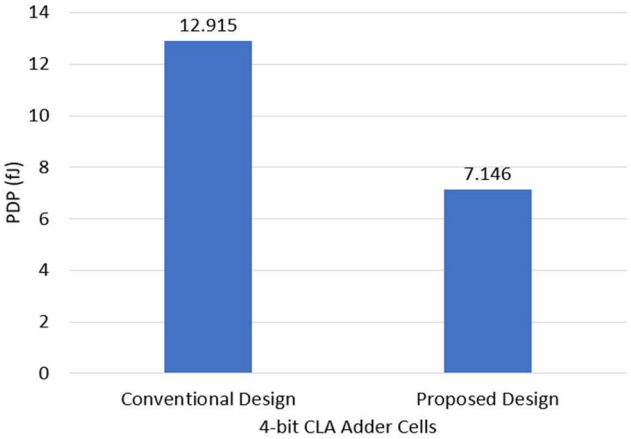


Fig. 7. PDP comparison.

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V. Comparative Analysis and Discussion

The conventional 4-bit CLA design only uses S-CMOS logic and transistor count for adder implementation is 170. The proposed design reduced transistor count by using GDI AND and XOR gates in [17]. Transistor count for the proposed design is 142. It resulted in less power (Fig. 5) and surface area consumption. Moreover, Using GDI gates in the input side reduced input impedance. It resulted in improvement in delay (Fig. 6). The low power and delay of the proposed design resulted in low PDP (Fig. 7) compared to the conventional design.

GDI technique-based logic gate does not have strong driving power like S-CMOS logic. However, using GDI gates in the proposed design does not bring about any change in the driving power of the CLA adder because GDI based gates are used in the input side. Output circuits for carry and sum generation are built with S-CMOS based circuits for which the proposed design would not lack driving power.

VI. Conclusion

A Hybrid 4-bit CLA adder using GDI and S-CMOS logic is presented in this research. The proposed design showed 35.28% improvement in average power and 14.371% improvement in delay. Due to improvement achieved in power and delay, the PDP of the proposed design obtained an improvement of 44.67%. Transistors required for the proposed design is also remarkably lower than the transistors of conventional design. Low transistor count accounts for low surface are. Therefore, the surface area on chip for the proposed 4-bit CLA adder implementation will also be lower than the conventional design. Hence, the design of circuits in this research for 4-bit CLA adder is highly suitable for building wide adder.

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